		velocity is halved.
23	**A**	$P_{device} = I^2R$ $4.0 = I^2 \times 20$ $I_{device} = 0.447$ A $P_{device} = I\ V$ $4.0 = 0.447\ V$ $V_{device} = 8.94$ V $V_{\mathrm{rheostat}} = I\ \mathrm{R}$ $16 - 8.94 = 0.447\ R_{rheostat}$ $R_{rheostat} = 15.7 \approx 16\ \Omega$
24	**C**	Given that the ammeter reading is zero, the p.d. across the 100 Ω is the same as